

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes three of the following points: - receptors on post-synaptic membrane (1) (receptors) have (different) three-dimensional shapes/ are specific for (different) neurotransmitters (1) - description of response once the neurotransmitter has bound to the receptor (1)	ACCEPT {ion channel / gates} on post-synaptic membrane ACCEPT (receptors have) complementary shape to specific neurotransmitter e.g. opening of sodium channels / causing an action potential / depolarising membrane / inhibitory neurotransmitter prevents {depolarisation /action potential}	(3)

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8(c)	An answer that includes two of the following points: • release neurotransmitters (1) • that act on white blood cells / (stimulate) {spleen/thymus/lymph nodes} (1) • to {release (more/ less) cytokines / release white blood cells into blood / recruit white blood cells from blood} (1)	ACCEPT release of named neurotransmitter e.g. noradrenaline / adrenaline / epinephrine / acetylcholine ignore release of chemicals ACCEPT release (more/ less) of named chemical e.g. histamine	(2)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes the following points: • record ECG before and (for 10 minutes) after changing {posture / position} (1) • count the number of a named peaks in one minute (1) • description of how POTS is identified (1)	ACCEPT calculate heart rate e.g. {heart rate / peaks per minute} will increase by 30 bpm / heart rate will be above 120 bpm (within 10 minutes of posture change)	(2)

Question number	Answer	Additional guidance	Mark
8(e)	An explanation that includes four of the following points: • bolus acts as a stimulus/ receptor detects bolus (1) • motor neurone transmits (electrical) impulse to (smooth) muscle (1) • detail of how nerve impulse stimulates muscle contraction (1) • therefore {circular muscle contracts / longitudinal muscles relax} {around/above} bolus (1) • therefore {circular muscle relax / longitudinal muscles constrict} in {front/below} of bolus (1)	ACCEPT description of food stretching gut wall ACCEPT release of neurotransmitter ACCEPT {circular muscle contracts / longitudinal muscles relax} in {propulsive / proximal} segment ACCEPT {circular muscle relax / longitudinal muscles constrict} in receiving segment	(4)

Question number	Answer	Additional guidance	Mark
8(f)	• inadequate blood supply to a {tissue / organ} (1)	ACCEPT a named organ e.g. heart ACCEPT {inadequate / decreased / no} blood supply	(1)

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8(g)	An answer that includes one of the following points: • mutation (in gene expressed by embryological ENS) (1) • (embryological ENS) do not have surface {antigens/ receptors} that allow them to locate to bowel (1)	ACCEPT so embryological ENS do not differentiate (into working neurones) ACCEPT conditions in the bowel may not be suitable for the growth of the embryonic cells	(1)

Question number	Answer	Additional guidance	Mark
8(h)	An explanation that includes three of the following points: • environmental factor changes conditions in the gut / causes a change in the gut flora (1) • therefore microbial {chemicals/toxins} released (1) • (chemicals / toxins / microbes) stimulate an {inflammatory response / immune response} (1) • explanation of how the inflammatory response occurs (1)	ACCEPT new microorganism / pathogen / microbe / bacteria ACCEPT release of {histamine / cytokines} e.g effect of histamine such as vasodilation / oedema occurs / recruitment of inflammatory cells / recruitment of white blood cells	(3)

